Victor S. Baidyshev

Curriculum Vitae

PERSONAL DATA

Affiliation Skolkovo Institute of Science and Technology, Moscow, Russia
Date of birth May 9, 1979
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RESEARCH INTERESTS

nanoparticles, gas-phase condensation, thermodynamic properties of nanoalloy, electron properties of nanoalloy, 2D material, DFT, MD, Machine learning

OTHER SKILLS

Scientific programs: Ovito, Origin, VESTA, Python, LAMMPS, DL_POLY, VASP, USPEX, DeePMD-kit

EMPLOYMENT and POSITIONS

12/2022 -present Skolkovo Institute of Science and Technology , Moscow, Russia
research
09/2009 -12/2022 Katanov Khakas State University, Abakan, Russia
assistant professor at the Department of Physics
11/2009 --06/2009 Kutateladze Institute of Thermophysics of the Siberian Branch of the Russian Academy of Sciences Novosibirsk, Russia
researcher
09/2004 -09/2009 Katanov Khakas State University, Abakan, Russia
researcher at the Nanophysics Laboratory

EDUCATION and DEGREES

10/2001-11/2004 Katanov Khakas State University Abakan, Russia
PhD student, 10.06.2005 - Candidate of Physico-Mathematical Sciences (Equivalent to Ph.D., Condensed Matter Physics). (Advisor: V. N. Udodov)
9/2005 – 8/2010 Katanov Khakas State University Abakan, Russia
Basic classical education, Physics

ADVANCED SCHOOLS

2/2014-3/2014 Forschungszentrum Jülich, Peter Grünberg Institute Jülich, Germany
45th IFF Spring School "Computing Solids: Models, Ab-initio Methods and Supercomputing"

GRANT HELD

1. Russian Science Foundation grant No. 22-23-20042, with parity financial support from the Government of the Republic of Khakassia, Bimetallic clusters promising for use in catalysis, Principal Investigator
2. Russian Science Foundation 21-73-20183 Design of low-dimensional nanomaterials for artificial photosynthesis, 2021 - 2024, performer
3. RFBR 17-42-190308 Complex experimental and theoretical study of epitaxial films of iron and manganese silicides, 2017-2018 performer, 2017-2018, performer
4. RFBR 17-48-190320 Development of experimental and theoretical foundations for the production of nanoparticles of various structures and chemical nature by the method of synthesis from a high-temperature gas environment, analysis of their physical and chemical properties, 2017 - 2018, performer

REFEREE FOR SCIENTIFIC JOURNALS

Nano Research, Journal of Catalysis, Applied Surface Science, Journal of Physical Chemistry C, Molecules, Membranes

PUBLICATIONS

1. Nikitin, D. S., Shanenkov, I. I., Nassyrbayev, A. R., Sivkov, A. A., Baidyshev, V. S., Kvashnina, Y. A., ... & Kvashnin, A. G. (2024). Synthesis of high-entropy Ti-Zr-Nb-Hf-Ta carbides and carbonitrides in high-speed arc discharge plasma jet. *Journal of Alloys and Compounds*, 177178. Kurenkova, A. Y., Radina, A. D., Baidyshev, V. S., Povalyaev, P. V., Aidakov, E. E., Gerasimov, E. Y., ... & Kvashnin, A. G. (2024). Photocatalytic H2 generation and CO2 reduction by WB5-x cocatalyst of TiO2 catalyst. *Applied Surface Science*, *661*, 160095.
2. Radina, A. D., Baidyshev, V. S., Chepkasov, I. V., Matsokin, N. A., Altalhi, T., Yakobson, B. I., & Kvashnin, A. G. (2024). Theoretical study of adsorption properties and CO oxidation reaction on surfaces of higher tungsten boride. *Scientific Reports*, *14*(1), 12788.
3. Chepkasov, I. V., Zamulin, I. S., Baidyshev, V. S., & Kvashnin, A. G. (2023). Tuning the surface properties of AuPd nanoparticles for adsorption of O and CO. *Physical Chemistry Chemical Physics*, *25*(48), 33031-33037.
4. Chepkasov, I. V., Baidyshev, V. S., & Kvashnin, A. G. (2023). Structure-driven tuning of O and CO adsorption on AuCu nanoparticles: A density functional theory study. *Physical Review B*, *108*(20), 205414.
5. Kvashnin, D. G., & Baidyshev, V. S. (2023). Study of formation and properties of F4TCNQ/MoS2 heterostructures with point defects driven by DFT and neural network potential. *FlatChem*, *42*, 100585.
6. Toksumakov, A. N., **Baidyshev, V. S**., Kvashnin, D. G., & Popov, Z. I. (2023). Bonding Duality and Optoelectronic Properties of Bilayer Carbon Structures Based on the T12 Phase and Penta-Graphene. *JETP Letters*, *117*(6), 441-448.
7. Leybo, D., Firestein, K. L., Evdokimenko, N. D., Ryzhova, A. A., **Baidyshev, V. S**., Chepkasov, I. V., ... & Shtansky, D. V. (2022). Ball-Milled Processed, Selective Fe/h-BN Nanocatalysts for CO2 Hydrogenation. *ACS Applied Nano Materials*, *5*(11), 16475-16488.
8. Sukhanova, E. V., **Baidyshev, V. S.**, Manakhov, A. M., Al-Qasim, A. S., & Popov, Z. I. (2023). Hydrogen production from H2S on metal-doped FeS Mackinawite monolayer via DFT calculations. *Applied Surface Science*, *609*, 155322.
9. Chepkasov, I. V., **Baidyshev, V. S**., Golubnichiy, A. A., Zamulin, I. S., Kvashnin, A. G., & Kozlov, S. M. (2022). Cu–Au nanoparticles produced by the aggregation of gas-phase metal atoms for CO oxidation: Special Issue: Emerging Investigators. *Aggregate*, *3*(6), e273.
10. Konopatsky, A. S., Kalinina, V. V., Savchenko, A. S., Leybo, D., Sukhanova, E. V., **Baidyshev, V. S**., ... & Shtansky, D. V. (2022). Structure, magnetic and adsorption properties of novel FePt/h-BN heteromaterials. Nano Research, 1-9.
11. Manakhov, A. M., Permyakova, E. S., Sitnikova, N. A., Tsygankova, A. R., Alekseev, A. Y., Solomatina, M. V., **V. S. Baidyshev**... & Solovieva, A. O. (2022). Biodegradable Nanohybrid Materials as Candidates for Self-Sanitizing Filters Aimed at Protection from SARS-CoV-2 in Public Areas. Molecules, 27(4), 1333.
12. Manakhov, A. M., Sitnikova, N. A., Tsygankova, A. R., Alekseev, A. Y., Adamenko, L. S., Permyakova, E., **V. S. Baidyshev**... & Solovieva, A. O. (2021). Electrospun biodegradable nanofibers coated homogenously by Cu magnetron sputtering exhibit fast ion release. Computational and experimental study. Membranes, 11(12), 965.
13. K. L. Firestein, N. D. Evdokimenko, A. L. Kustov, **V. S. Baidyshev**, I. V. Chepkasov, Z. I. Popov, A. T. Matveev, I. V. Shetinin, D. V. Leybo, I. N. Volkov, A. M. Kovalsky, D. Golberg, D. V. Shtansky Microstructure and catalytic properties of Fe3O4/BN, Fe3O4 (Pt)/BN, and FePt/BN heterogeneous nanomaterials in CO2 hydrogenation reaction: Experimental and theoretical insights //Journal of Catalysis. – 2021. – T. 402. – C. 130-142.
14. Chepkasov I.V**.**, **Baidyshev V.S**., Sukhanova E.V., Visotin M.A., Süle P., Popov Z.I. Iron silicides formation on Si (100) and (111) surfaces through theoretical modeling of sputtering and annealing // Applied Surface Science. – 2020. – C. 146736.
15. Chepkasov I. V**.**, Visotin M. A., Kovaleva E. A., Manakhov A. M., **Baidyshev V. S**., Popov Z. I. Stability and Electronic Properties of PtPd Nanoparticles via MD and DFT Calculations //The Journal of Physical Chemistry C. – 2018. – V. 122. – №. 31. – P. 18070-18076.
16. **Baidyshev V. S.,** Chepkasov I. V., Artemova N. D. Study of thermal stability of disordered alloy AgxCu1-x nanoparticles by molecular dynamic simulations //Journal of Physics: Conference Series. 2018. – V. 1015. – №. 3. – P. 032021.
17. Chepkasov I. V., **Baidyshev V. S.,** Baev A. Y. Structural properties of CuAu nanoparticles with different type. Molecular dynamic simulations //Journal of Physics: Conference Series. 2018. – V. 1015. – №. 3. – P. 032022.
18. Chepkasov I. V., **Baidyshev V. S.,** Tsura V. A. Molecular dynamic simulation of melting copper-silicon nanoparticles //Journal of Physics: Conference Series. 2018. – V. 1015. – №. 3. – P. 032023.
19. **Baidyshev, V. S**., Gafner, Y., Gafner, S. L., & Redel, L. V. (2017). Thermal stability of Pt nanoclusters interacting to carbon sublattice. Physics of the Solid State, 59(12), 2512-2518.

20. **Baidyshev V. S**. et al. Formation of structure in small lead clusters under thermal effect //Crystallography Reports. – 2016. – Т. 61. – №. 7. – С. 1096-1101.
21. Gafner, Y., Gafner, S. L., Zamulin, I. S., Redel, L. V., **& Baidyshev, V. S**. (2015). Analysis of the heat capacity of nanoclusters of FCC metals on the example of Al, Ni, Cu, Pd, and Au. *The Physics of Metals and Metallography*, *116*(6), 568-575.
22. Gafner, Y. Y., & **Baidyshev, V. S.** (2015, March). The role of the vacancy complexes of nanocompacted aluminum particles on the formation of structure during heat treatment. In IOP Conference Series: Materials Science and Engineering (Vol. 81, No. 1, p. 012012). IOP Publishing.
23. **Baidyshev, V. S.,** Gafner, Y., Samsonov, V. M., & Bembel, A. G. (2015). Modeling possible structural transitions in aluminum and lead nanoclusters. Crystallography Reports, 60(1), 95-100.
24. Gafner, Y., **Baidyshev, V. S**., & Gafner, S. L. (2015). Effect of disorder on the structure of small aluminum clusters during heat treatment. Physics of the Solid State, 57(1), 188-196.